



National Treasury Secure Data Facility

Safe output rules for researchers

December 2024 (version 1)

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Introduction

- The National Treasury – Secure Data Facility (NT-SDF) hosts de-identified tax data from the South African Revenue Service (SARS), alongside selected datasets from other sources. Research conducted with these data must adhere to stringent security protocols that meet both the requirements of the data suppliers and the NT-SDF's own security standards.
- Following training in data management and engagements with other secure labs, the NT-SDF has adopted the *five-safes* approach to data security. *Safe output* governs the rules by which researchers can remove information from the secure facility. The aim is to prevent unwanted disclosure of sensitive information, and we refer to this practice as Statistical Disclosure Control (SDC).
- This document outlines the output rules adopted by the NT-SDF. It is important to recognize that even very comprehensive rules cannot account for every possible type of output request. Therefore, each request will be evaluated not only against these rules but also with a common-sense approach that considers the broader context and potential implications of releasing the output. Consequently, the NT-SDF may, at times, reject outputs that technically comply with the rules if broader considerations warrant such a decision.

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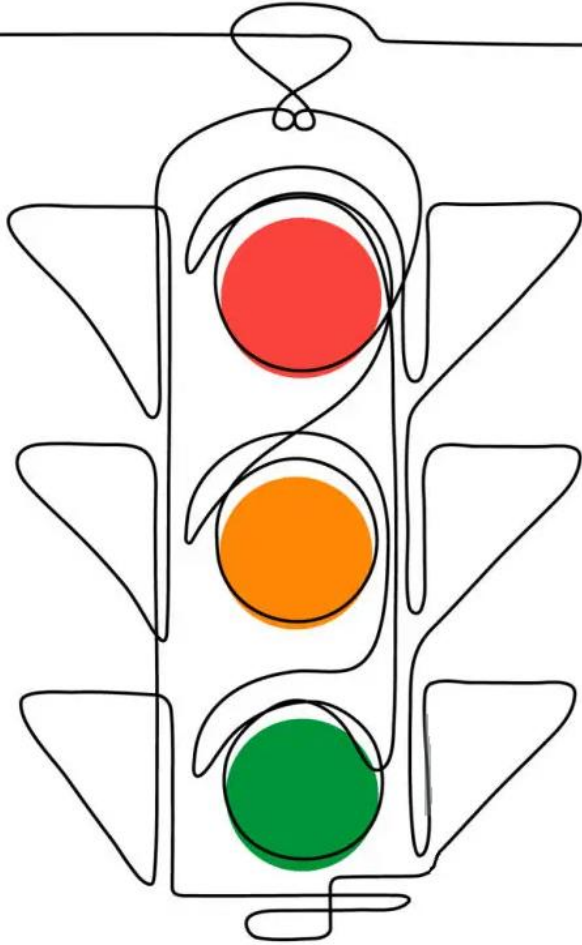
A stylized landscape illustration featuring a green field in the foreground, a city skyline with a large orange sun in the background, and three wind turbines on the right. The sky is a gradient of yellow and orange.

What can leave the NT-SDF?

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What can leave the NT-SDF



NEVER

Row-level data, counts less than 10, ID information, disclosure of any kind, information unrelated to project, output not checked by the NT-SDF data team

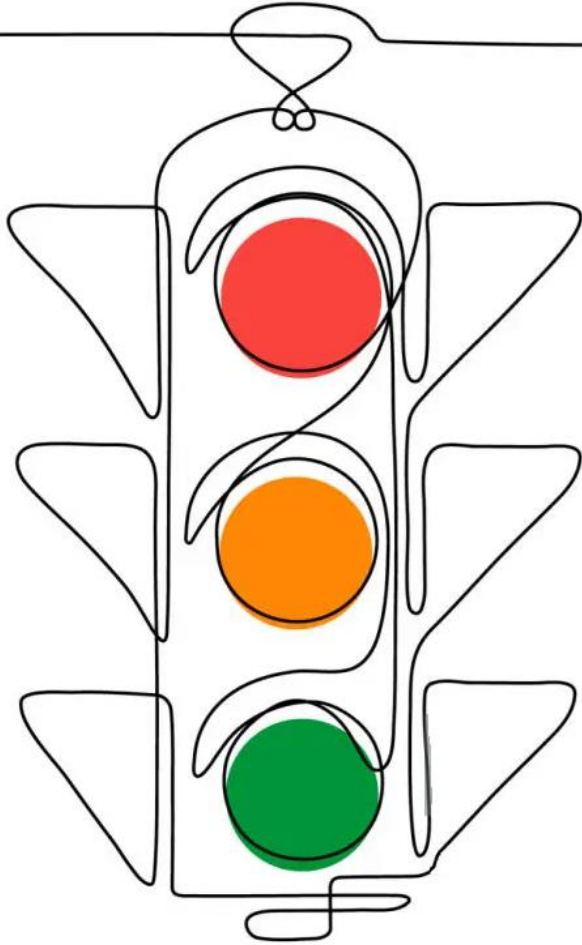
WITH PERMISSIONS AND SPECIAL CHECKS

Geographic analysis, industry analysis, aggregate tables, information created outside of your project (someone else's code/results)

ALLOWED – UNIVERSAL CHECKS ONLY

Regression and statistical results, results in highly aggregated tables, code

File formats for in and out files



NEVER

Project (rproj), executables (exe, bat), certain graph types (eps, gph), unusual or unknown file formats

SPECIAL CHECKING

Python syntax files (.pynb) , datafiles being brought in

ALLOWED – UNIVERSAL CHECKS ONLY

Image (png, jpg), statistical results (smcl, tex, docx), log (log, txt, smcl), tables (xlsx, dta), code (do, py, r), documents (pdf, docx)

A stylized landscape illustration featuring a green field in the foreground, a city skyline with various buildings and a prominent dome in the middle ground, and rolling hills in the background. The sky is a gradient of yellow and orange, suggesting a sunset or sunrise. Wind turbines are visible on the right side of the horizon.

Output rules

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Output rules – all rules



The rule of 10

All tabular and similar output should have at least 10 units (unweighted) underlying any cell or data point presented. Any statistics derived from less than 10 data subjects (zero to nine inclusive) must be removed. Counts at the data subject level must be provided for every output, including safe outputs (e.g. regression coefficients).

The rule of 10 – degrees of freedom

All modelled output should have at least 10 degrees of freedom.

Class disclosure

Sensitive information cannot be disclosed about groups. This will be evaluated on a case-by-case basis, depending on group size, uniqueness and the information being removed. (See later slide that contains an example).



Dominance disclosure

In all tabular and similar data, the largest contributor of a cell cannot exceed 50% of the cell total to prevent dominance disclosure. This will be evaluated on a case-by-case basis. (See later slide that contains an example).

Non-reversibility of SDC

SDC must not be reversible (for example, by differencing). If SDC is reversible then secondary SDC must be applied (such as secondary suppression or reformatting). (See later slide that contains an example).

Results only

Output should 'look like' output and not like data. Output is for the purposes of inclusion in academic work and not for the purposes of further analysis. As a rule of thumb, the size of an A4 page.



Lowest geo level

The threshold for spatial aggregation is the municipal level. No spatial aggregation can occur at a lower level than this (e.g. suburb) without special permission*.

Lowest industry level

The threshold for industry aggregation is the 1-digit industry code. No industry aggregation can occur at a lower level than this (e.g. 2-digit, 3-digit industry code etc.) without special permission*.

File formats

Stick to recommended formats and avoid formats that are not allowed (rproj, exe, bat, eps, gph, unusual or unknown file formats).



Concentration ratios

Concentration ratios must not include decimal places.

Residuals

Information on residuals must be aggregated to the sample level and satisfy all other rules. Plots of residuals need to be aggregated.

Factor scores and proxy variables

Factor scores and summary information on factor scores must be related to at least two different variables. Proxy variables must be explained.



Output is understandable

The output must be self-explanatory so that rules can be applied. Unclear output will not be released. No output released which is outside of the project scope.

No ID information

Hashed IDs cannot be removed from the lab, especially in high volumes. Syntax files referring to one or two hashed IDs may be permitted.

Graphs

Graphs based on actual data points must aggregate data points (e.g. histogram). Graphs based on aggregated information (coefficients, statistics) can be disaggregated (eg scatter plot)

Output rules - by statistic/ variable

Risk	Statistic type	Rules
	Correlation coefficients	Universal Rules
	Moments of distributions	
	Counts/ percentages	
	Means	
	P-shares, P-ratios	
	Summary and test statistics	
	Weighted counts/ percentages	
	Dataset summaries	ID information not included
	Concentration ratios	No decimal places
High	Regression	No disaggregated residuals
	Percentiles	100 percentiles max per request
	Maximum, Minimum, Mode	Not allowed in aggregate tables
	Aggregate tables	Must be results based
High	Industry	Lowest industry level is 1-digit*
	Geo data	Lowest geo level is municipality*

Universal rules:

- Results only
- The rule of 10
- No class or dominance disclosure
- Non-reversibility of SDC
- Nothing outside scope
- Output must be understandable

Notes:

- Items marked with an asterisk can be negotiated for special permissions during the proposal stage

A stylized landscape illustration at the top of the slide. It features rolling hills in shades of yellow and orange. In the middle ground, there is a city skyline with various buildings, including a prominent dome. To the right of the city, there are three wind turbines. The background is a bright yellow sky with a large orange sun. The foreground is a solid dark green field.

Understanding disclosure

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Concepts and definitions

- Understanding disclosure is easiest with examples, which we'll provide below. However, it helps to first define what we mean by disclosure. In our context, disclosure is:
- “When sensitive information about a statistical unit (e.g., an individual or a firm) or group of units is revealed, even though it is intended to remain confidential.”
- Consider the following key points:
- **Sensitive Information:** This can be subjective. For instance, is a person's income considered sensitive? We generally err on the side of caution, especially with income data, which is a significant part of tax data.
- **Unit or Group of Units:** Statistical Disclosure Control (SDC) focuses on preventing the identification of individual units. However, revealing sensitive insights about a group can also constitute disclosure.
- Lastly, these concepts often interact. For example, a study on geography or industry alone may be permissible, but combining industry and geographic data might be considered too sensitive. Therefore, in addition to formal rules, the NT-SDF takes a pragmatic, case-by-case approach, applying principles to assess each situation.

Dominance disclosure problem – firm example

Sector	Firm Size	Total employees	Firm count	Total Revenue (in R mil)
Water	Small	452	30	10
Water	Medium	811	11	211
Water	Large	1386	10	345
Electricity	Small	2000	15	70
Electricity	Medium	5000	30	100
Electricity	Large	41199	10	256 000

Revenue is often seen as a sensitive variable and the dominance in this example could help us estimate the firm's revenue

This cell is dominated by a potentially well-known large firm

Dominance without potential disclosure – firm example

Sector	Firm Size	Total employees	Firm count
Water	Small	452	30
Water	Medium	811	11
Water	Large	1386	10
Electricity	Small	2000	15
Electricity	Medium	5000	30
Electricity	Large	41199	10

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This cell is dominated by a potentially well-known large firm

If total employment is not considered a sensitive variable, the dominance might not be a problem.

Class disclosure* - suburb example

Area	Average income	Max income	Count
Glendale	50000	400000	3000
Ruworth	500	1000	1000
Shifford	20000	10000	5000
Melville	30000	200000	6000
Brigpond	120000	500000	4000
Littleview	100000	200000	2000

Here we can learn that everyone in Ruworth earns less than 1000.

If the variable or value is not sensitive, this may not be a problem

* Note: Class disclosure is sometimes called attribute disclosure, or group disclosure.

Example of Reversible SDC

Age group	Gender	Count	Mean income
20-40	Male	502	10000
20-40	Female	451	8000
40-60	Male	390	20000
40-60	Female	380	20000
60-80	Male	90	50000
60-80	Female	96	45000
81+	Male	14	100000
81+	Female	(value removed as N<10)	120000
Total		1928	

The researcher attempted to accommodate the rule of 10 by redacting the cell containing a value less than 10. However, due to the inclusion of the column total at the bottom, it becomes possible to reverse-engineer this redacted value.

This issue arises when 'Total' columns or rows are present, and a problematic cell is redacted.

For the output to be released, the 'Total' column must be removed to ensure compliance with the rule of 10.

A stylized landscape illustration featuring a green field in the foreground, a city skyline with various buildings and a large orange sun in the background, and rolling hills under a yellow sky. Wind turbines are visible on the right side of the horizon.

Requesting output

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Output request template

Dear Data Team, I would like to request the following output:	
Project ID/ Workbench name	2024.tax.doe.1
Folder path and name	John_doe\out_files\2024.10.20 [etc etc]
Contents	5 regression tables of age and industry on wage
Special permissions	This project has been permitted to go to the 2-digit level of industry codes.
Syntax/do file/script only request	No
Regards, [researcher name]	

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*Have this structure
in your email*

ID assigned to project, or workbench name if ID not available

Specify the file path and the name of the folder you are requesting. Please prefix the folder name with the date in *yyyy.mm.dd* format.

Briefly describe the contents

If your output is part of a project that has been granted special permissions, please note that here.

The expected output return time is 7 working days. Only code related files can be expedited.



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